Notes on Managing with Style: The Effect of Managers on Firm Policies

Bertrand & Schoar (QJE, 2003)

Contents

1 Big picture and research question

These notes distill the empirical framework and main results in Bertrand and Schoar (2003), which asks a simple but deep question:

> *Do individual top managers leave a "fingerprint" on firm policies and performance beyond what can be explained by firm and year effects?*

The paper is positioned between (i) corporate finance/organizational economics theories where managers matter, and (ii) empirical work that typically treats firms as the unit of analysis. The key innovation is to build a *manager–firm matched panel* that tracks the same executive across *different firms over time.* This mobility is essential: it creates within-manager variation across firm environments, and within-firm variation across different managers.

The empirical contribution is therefore *not* to claim random assignment of managers to firms. Instead, the paper documents whether policies *systematically covary with manager identity* once we partial out firm and year effects and standard firm-level controls.

2 Data construction and variables

2.1 Manager–firm matched panel

The sample is built by matching top executives (from ExecuComp) to firm-year observations (from Compustat) and acquisition activity (from SDC). Identification relies on *manager "switchers"*: managers observed in multiple firms. To avoid confounding manager fixed effects with time-varying firm shocks, the paper focuses on managers who spend *at least three years* in each firm they are observed in.

Intuitively, this yields a panel of large publicly listed U.S. firms in which some executives are observed moving across firms and, often, across industries.

2.2 Variable definitions (paraphrased)

The corporate policy and performance variables are constructed from standard accounting and deal-flow items. In these notes, it is useful to keep the following broad mapping in mind:

- **Investment intensity**: capital expenditures scaled by beginning-of-year net PPE.
- **Tobin's Q**: market value of assets divided by book assets (constructed from Compustat components).
- **Cash flow**: earnings before extraordinary items plus depreciation, scaled by beginning-of-year net PPE.
- **Capital structure & liquidity**: leverage, cash holdings, and interest coverage.

- **Payout policy**: dividends scaled by earnings.
- **Organizational strategy**: acquisition counts, diversifying acquisitions, R&D intensity, advertising intensity, and SG&A.
- **Performance**: return on assets and operating return on assets.

Table 1 provides descriptive statistics and highlights that the manager-matched sample consists of very large firms relative to the Compustat universe.

3 Core empirical framework: manager fixed effects

3.1 Baseline specification

For each outcome y_{it} (a corporate policy or performance metric for firm i in year t), the paper estimates a fixed-effects panel regression of the form:

$$y_{it} \;=\; \alpha_t \;+\; \alpha_i \;+\; X_{it}'\beta \;+\; \delta^{\text{CEO}}_{m(i,t)} \;+\; \delta^{\text{CFO}}_{m(i,t)} \;+\; \delta^{\text{Other}}_{m(i,t)} \;+\; \varepsilon_{it}. \tag{1}$$

Objects in (1)**.**

- α_i: firm fixed effects (time-invariant firm heterogeneity).
- α_t: year fixed effects (aggregate shocks).
- X_{it}: time-varying firm controls (chosen per dependent variable; see the notes under Tables 3–4).
- $\delta^{\text{CEO}}, \delta^{\text{CFO}}, \delta^{\text{Other}}$: manager fixed effects for executives categorized by their *final observed role* (CEO, CFO, or other top executive).

Interpretation. A statistically significant set of manager fixed effects means that, after controlling for firm identity, time, and observables, the outcome differs systematically when different managers are in charge.

Estimation details. Standard errors are clustered at the firm level in the baseline tables.

3.2 Manager effects on sensitivities (investment–Q and investment–cash flow)

Beyond level effects (e.g., investment intensity), the paper also studies whether managers differ in *how* they map signals into decisions. The most prominent examples are:

- **Investment–Q sensitivity**: how strongly investment responds to Tobin's Q.
- **Investment–cash flow sensitivity**: how strongly investment responds to internal cash flow.

Operationally, these are estimated by interacting the relevant fixed effects with $Q_{i,t-1}$ or cash flow. A stylized version is:

$$
\begin{aligned}
\text{Investment}_{it} = \alpha_t + \alpha_i + X_{it}'\beta + \sum_m \theta_m^{\text{CEO}} \nVdash\{m = \text{CEO}(i,t)\} \cdot Q_{i,t-1} + \sum_m \theta_m^{\text{CFO}} \nVdash\{m = \text{CFO}(i,t)\} \cdot Q_{i,t-1} \\
+ \sum_m \theta_m^{\text{Other}} \nVdash\{m = \text{Other}(i,t)\} \cdot Q_{i,t-1} \qquad (2) \\
+ \text{(analogous interactions with cash flow for CF sensitivity)} + \varepsilon_{it}. \qquad (3)
\end{aligned}
$$

The F-tests reported for these sensitivity outcomes are tests of the joint significance of the interaction terms (manager FE $\times$ Q or manager FE $\times$ cash flow).

4 Do managers matter for policies? Evidence from fixed effects tests

Tables 3 and 4 are the backbone of the paper. They report, for each outcome, a sequence of specifications:

1. Firm FE + year FE only.
2. Add CEO fixed effects.
3. Add CEO + CFO + other executive fixed effects.

The reported statistics are F-tests for the joint significance of the relevant manager fixed effects and the adjusted R^2.

4.1 Investment and financial policies (Table III)

Main reading of Table 3.

- Manager fixed effects are strongly statistically significant for many policies.
- CEO fixed effects matter for investment levels, investment–Q sensitivity, acquisitions, cash holdings, and dividends.
- CFO fixed effects are especially strong for investment levels and investment–Q sensitivity, and also appear in leverage, interest coverage, and cash holdings.
- "Other" executives matter in several outcomes, particularly M&A-related measures.

The pattern is consistent with the idea that (i) CEOs matter broadly, (ii) CFOs matter particularly for financial choices and the mapping from financial signals to investment, and (iii) operational executives matter for specific strategic actions such as acquisitions.

4.2 Organizational strategy and performance (Table IV)

Main reading of Table 4.

- Manager fixed effects are significant not only for policies but also for *performance* (ROA).
- CEO effects are very strong for SG&A (a proxy for cost structure/organization).
- For operating ROA, the evidence is weaker for CFOs and other executives in the most saturated specification, suggesting that some performance measures are more tightly linked to CEO-level styles (or are noisier once many fixed effects are included).

5 Are these effects persistent? Robustness via "across-jobs" regressions

A key concern is whether estimated manager fixed effects merely reflect coincidence with a firm-specific shock (e.g., a manager happened to be present during an acquisition wave). The paper therefore tests *persistence across firms*.

5.1 Persistence regression (Table V)

The logic is:

1. First residualize each policy y_{it} using firm FE, year FE, and controls to obtain $\hat{u}_{it}$.
2. Collapse residuals to the manager–firm-spell level (average within a manager's tenure at a firm).
3. Regress the average residual in the manager's *second* observed firm on the average residual in the *first* observed firm:
$$\bar{u}_m^{(2)} = a + b\,\bar{u}_m^{(1)} + \eta_m. \tag{4}$$

Table 5 reports b (with standard errors and R^2) for many outcomes and shows strong positive persistence in the real data, while a placebo timing test produces coefficients close to zero.

6 How large are manager effects? Distributional magnitudes

Statistical significance does not automatically imply economic importance. Table 6 summarizes the *distribution* of estimated manager fixed effects (weighted to account for estimation error).

Main reading. Comparing the 25^{th} and 75^{th} percentiles yields economically meaningful gaps in investment, acquisitions, leverage, cash holdings, and performance.

7 From isolated effects to "styles": correlation patterns across policies

Once manager fixed effects exist for many policies, the next step is to ask whether they form *coherent patterns*—i.e., *styles*. The paper examines cross-policy associations by regressing one set of manager fixed effects on another:

$$\widehat{\delta}_m^{(A)} = a + b\,\widehat{\delta}_m^{(B)} + \xi_m, \tag{5}$$

using weighted least squares (weights based on estimation precision).

Table 7 reports these pairwise relationships and highlights significant coefficients.

Examples of economic interpretation from Table 7.

- **Investment benchmark substitution:** managers with higher investment–Q sensitivity tend to have lower investment–cash flow sensitivity ($b < 0$).
- **Financial aggressiveness:** higher leverage fixed effects are associated with lower cash holdings fixed effects.
- **Cost discipline and performance:** higher ROA fixed effects are associated with substantially lower SG&A fixed effects.

8 Governance and compensation: are "better styles" rewarded?

The paper then asks whether performance-related styles correlate with governance and pay.

8.1 Governance

For governance, the outcome is the manager fixed effect (from Tables 3–4) and the regressor is a blockholder concentration measure in the manager's *second* firm:

$$\widehat{\delta}_m^{(A)} = a + b\,\mathrm{BlockholderShare}_{\mathrm{firm2}(m)} + \nu_m. \tag{6}$$

Column (1) of Table 8 suggests that managers with higher performance fixed effects are more likely to be found in firms with more concentrated ownership.

8.2 Compensation

To isolate pay unrelated to firm effects and standard observables, the paper constructs manager-level compensation residuals. A stylized first-stage regression is:

$$\log(\mathrm{Pay}_{it}) = \alpha_i + \alpha_t + Z_{it}'\gamma + \mathrm{RoleDummies} + \mathrm{Tenure} + e_{it}, \tag{7}$$

where Z_{it} includes size controls and profitability controls. Residuals from (7) are then collapsed to the manager level and regressed on manager fixed effects:

$$\text{ResidualPay}_m = a + b\,\widehat{\delta}_m^{(A)} + \omega_m. \tag{8}$$

Columns (2) and (3) of Table 8 show that managers with higher ROA fixed effects earn higher residual compensation, consistent with boards paying a premium for high-performing styles.

9 Observable CEO traits: cohort effects and MBA

The manager fixed-effects results naturally raise a follow-up question: *can we link some of this heterogeneity to observable managerial characteristics?* The paper focuses on two CEO traits that are widely available in ExecuComp:

- **Birth cohort** (proxied by CEO year of birth).
- **MBA indicator** (equals 1 if the CEO holds an MBA).

9.1 Equation (3): levels of policies and performance

Let i index firms, j index CEOs, and t index years. For most outcomes, the paper estimates:

$$y_{ijt} \;=\; X_{it}'\beta \;+\; \gamma\,\text{MBA}_j \;+\; \phi\,\text{Cohort}_j \;+\; \kappa\,\text{Tenure}_{j,t} \;+\; \alpha_i \;+\; \alpha_t \;+\; \varepsilon_{ijt}. \tag{9}$$

Here X_{it} contains firm controls (which vary by dependent variable), α_i and α_t are firm and year fixed effects, and CEO tenure is included to capture potential learning/entrenchment dynamics. Standard errors are clustered at the CEO (manager) level.

9.2 Equation (4): investment sensitivities

To study how CEO traits map into *investment–cash flow* and *investment–Q* sensitivities, the paper uses a richer specification that allows (i) trait-specific sensitivities and (ii) baseline firm-specific sensitivities. A transparent way to write their approach is:

$$\begin{aligned} I_{ijt} = {} & X_{it}'\beta + \gamma_1\text{MBA}_j + \gamma_2\text{MBA}_j \cdot \frac{CF_{it}}{K_{i,t-1}} + \gamma_3\text{MBA}_j \cdot Q_{i,t-1} \\ & + \phi_1\text{Cohort}_j + \phi_2\text{Cohort}_j \cdot \frac{CF_{it}}{K_{i,t-1}} + \phi_3\text{Cohort}_j \cdot Q_{i,t-1} \\ & + \kappa_1\text{Tenure}_{j,t} + \kappa_2\text{Tenure}_{j,t} \cdot \frac{CF_{it}}{K_{i,t-1}} + \kappa_3\text{Tenure}_{j,t} \cdot Q_{i,t-1} \\ & + \alpha_i + \alpha_t + \underbrace{\alpha_{i,CF} \cdot \frac{CF_{it}}{K_{i,t-1}} + \alpha_{i,Q} \cdot Q_{i,t-1}}_{\text{firm-specific baseline sensitivities}} + \varepsilon_{ijt}. \end{aligned} \tag{10}$$

The coefficients γ_3 and ϕ_3 capture how MBA status and cohort shift investment–Q sensitivity, while γ_2 and ϕ_2 capture how those traits shift investment–cash flow sensitivity.

Reading Table 9. Table 9 reports the cohort and MBA coefficients for many policies (from (9)) and, for the sensitivity measures, the relevant interaction coefficients from (10). The estimates suggest that younger cohorts (higher birth year) adopt more aggressive financial policies, while MBA CEOs tend to rely more on Q and less on internal cash flow when making investment decisions.

10 Tables (reproduced) and detailed interpretation

10.1 Table I: Descriptive statistics

Table 1: Descriptive statistics

	Manager–firm matched sample		Manager characteristics sample		Compustat	
Variable	Mean	Std. dev.	Mean	Std. dev.	Mean	Std. dev.
Total sales	5606.5	11545.6	5333.3	10777.4	2649.6	5878.2
Investment	0.39	2.94	0.28	0.50	0.34	2.67
Average Tobin's Q	2.40	3.85	2.03	2.05	1.70	1.43
Cash flow	0.44	1.91	0.45	2.10	0.43	2.47
Number of acquisitions	0.77	1.48	0.65	1.40	0.36	1.45
Leverage	0.35	0.39	0.34	0.28	0.45	1.21
Interest coverage	35.0	875.1	40.5	663.1	27.6	166.2
Cash holdings	0.11	0.16	0.08	0.11	0.17	0.80
Dividends/earnings	0.11	0.79	0.14	1.05	0.16	0.25
Diversifying acquisitions	0.32	1.09	0.28	0.91	0.12	0.63
R&D	0.05	0.07	0.04	0.14	0.03	0.06
Advertising	0.05	0.06	0.05	0.06	0.04	0.06
SG&A	0.26	0.98	0.21	0.19	0.18	0.64
Return on assets	0.16	0.11	0.19	0.15	0.10	0.09
Operating return on assets	0.09	0.12	0.11	0.22	0.08	0.13
Sample size	6766		10472		38489	

Notes: The manager–firm matched sample contains firm-years for firms that employ at least one manager observed in multiple firms with a minimum three-year stay per firm. The manager characteristics sample further restricts to observations with CEO birth year and MBA information.

Interpretation. The matched sample consists of large firms (mean sales above \$5B) with relatively high Tobin's Q and active acquisition behavior compared with the broader Compustat sample. This matters because the paper's inferences are about the segment of large public firms where top-manager mobility is observed.

10.2 Table II: Executive transitions

Interpretation. Two facts are central for identification. First, there is substantial mobility (hundreds of across-firm moves). Second, many moves cross industries (e.g., 63% of CEO→CEO moves are cross-industry), suggesting that what is transported is not purely industry-specific operational knowledge but *potentially general managerial styles*.

Table 2: Executive transitions between positions and industries

	CEO	CFO	Other
from CEO	117 (63%)	4 (75%)	52 (69%)
from CFO	7 (71%)	58 (71%)	30 (57%)
from Other	106 (60%)	0 (–)	145 (42%)

Notes: Each cell reports the number of across-firm transitions from the row position to the column position. The percentage in parentheses is the fraction of those transitions that cross two-digit industries.

Table 3: Executive effects on investment and financial policies (F-tests on manager fixed effects)

Panel	Dependent variable	Spec.	F_{CEO}	F_{CFO}	F_{Other}	N	Adj. R^2
		(1) Firm+year FE	–	–	–	6631	0.91
	Investment	(2) + CEO FE	16.74 (< 0.0001, 198)	–	–	6631	0.94
		(3) + CEO+CFO+Other FE	19.39 (< 0.0001, 192)	53.48 (< 0.0001, 55)	8.45 (< 0.0001, 200)	6631	0.96
		(1) Firm+year FE	–	–	–	6631	0.95
	Inv. to Q sensitivity	(2) + CEO FE	17.87 (< 0.0001, 223)	–	–	6631	0.97
A: Investment policy		(3) + CEO+CFO+Other FE	5.33 (< 0.0001, 221)	9.40 (< 0.0001, 58)	20.29 (< 0.0001, 208)	6631	0.98
		(1) Firm+year FE	–	–	–	6631	0.97
	Inv. to CF sensitivity	(2) + CEO FE	2.00 (< 0.0001, 205)	–	–	6631	0.98
		(3) + CEO+CFO+Other FE	0.94 (0.7276, 194)	1.29 (0.0760, 55)	1.28 (0.0058, 199)	6631	0.98
		(1) Firm+year FE	–	–	–	6593	0.25
	Number of acquisitions	(2) + CEO FE	2.01 (< 0.0001, 204)	–	–	6593	0.28
		(3) + CEO+CFO+Other FE	1.68 (< 0.0001, 199)	1.74 (0.0006, 55)	4.08 (< 0.0001, 203)	6593	0.36
		(1) Firm+year FE	–	–	–	6563	0.39
	Leverage	(2) + CEO FE	0.99 (0.5294, 203)	–	–	6563	0.39
		(3) + CEO+CFO+Other FE	0.86 (0.9190, 199)	1.43 (0.0225, 54)	1.21 (0.0230, 203)	6563	0.41
		(1) Firm+year FE	–	–	–	6278	0.31
	Interest coverage	(2) + CEO FE	0.56 (0.99, 193)	–	–	6278	0.31
B: Financial policy		(3) + CEO+CFO+Other FE	0.35 (0.99, 192)	13.85 (< 0.0001, 50)	2.61 (< 0.0001, 192)	6278	0.41
		(1) Firm+year FE	–	–	–	6592	0.77
	Cash holdings	(2) + CEO FE	2.52 (< 0.0001, 204)	–	–	6592	0.78
		(3) + CEO+CFO+Other FE	2.48 (< 0.0001, 201)	3.68 (< 0.0001, 54)	2.53 (< 0.0001, 202)	6592	0.80
		(1) Firm+year FE	–	–	–	6580	0.65
	Dividends/earnings	(2) + CEO FE	5.78 (< 0.0001, 203)	–	–	6580	0.71
		(3) + CEO+CFO+Other FE	4.95 (< 0.0001, 199)	1.07 (0.3368, 54)	1.74 (< 0.0001, 203)	6580	0.72

Notes: Each row corresponds to a fixed-effects panel regression. Standard errors are clustered at the firm level. Sensitivity outcomes (investment–Q, investment–cash flow) are identified using interactions of fixed effects with lagged Q or cash flow, respectively.

Table 4: Executive effects on organizational strategy and performance (F-tests on manager fixed effects)

Panel	Dependent variable	Spec.	F_{CEO}	F_{CFO}	F_{Other}	N	Adj. R^2
A: Organizational strategy	Diversifying acquisitions	(1) Firm+year FE	–	–	–	6593	0.22
		(2) + CEO FE	2.06 (< 0.0001, 204)	–	–	6593	0.25
		(3) + CEO+CFO+Other FE	1.23 (0.0163, 202)	1.74 (0.0007, 53)	3.97 (< 0.0001, 202)	6593	0.33
	R&D	(1) Firm+year FE	–	–	–	4283	0.78
		(2) + CEO FE	1.86 (< 0.0001, 145)	–	–	4283	0.79
		(3) + CEO+CFO+Other FE	2.27 (< 0.0001, 143)	3.60 (< 0.0001, 45)	4.46 (< 0.0001, 143)	4283	0.83
	Advertising	(1) Firm+year FE	–	–	–	2584	0.79
		(2) + CEO FE	2.88 (< 0.0001, 95)	–	–	2584	0.81
		(3) + CEO+CFO+Other FE	4.03 (< 0.0001, 95)	0.84 (0.6665, 21)	6.10 (< 0.0001, 80)	2584	0.84
	SG&A	(1) Firm+year FE	–	–	–	2397	0.46
		(2) + CEO FE	33.55 (< 0.0001, 123)	–	–	2397	0.83
		(3) + CEO+CFO+Other FE	13.80 (< 0.0001, 118)	0.82 (0.7934, 42)	0.77 (0.9777, 146)	2397	0.83
B: Performance	Return on assets	(1) Firm+year FE	–	–	–	6593	0.72
		(2) + CEO FE	2.04 (< 0.0001, 217)	–	–	6593	0.74
		(3) + CEO+CFO+Other FE	2.46 (< 0.0001, 201)	3.39 (< 0.0001, 54)	4.46 (< 0.0001, 202)	6593	0.77
	Operating return on assets	(1) Firm+year FE	–	–	–	5135	0.34
		(2) + CEO FE	2.61 (< 0.0001, 217)	–	–	5135	0.39
		(3) + CEO+CFO+Other FE	1.60 (< 0.0001, 216)	0.66 (0.9788, 58)	1.01 (0.4536, 217)	5135	0.39

Notes: Each row corresponds to a fixed-effects panel regression with standard errors clustered at the firm level. Controls differ by outcome (size, profitability, cash flow, and acquisition dummies as appropriate).

10.3 Table III: Executive effects on investment and financial policies

Interpretation. The F-tests reject the null of no manager effects for many outcomes. Economically, this means the identity of the CEO/CFO is associated with systematically different policy choices even within the same firm over time.

10.4 Table IV: Executive effects on organizational strategy and performance

Interpretation. Manager effects appear not only for policies but also for performance. This strengthens the interpretation that "style" is not simply cosmetic: it correlates with outcomes that are economically consequential.

10.5 Table V: Persistence of manager effects

Table 5: Persistence of manager effects: real data vs. placebo

	Real data	Placebo data
Investment	0.05 (0.02) [0.01]	0.01 (0.02) [0.00]
Number of acquisitions	0.49 (0.05) [0.13]	−0.02 (0.05) [0.00]
Leverage	0.40 (0.03) [0.21]	0.02 (0.05) [0.01]
Cash holdings	0.74 (0.05) [0.35]	0.05 (0.07) [0.01]
Dividends/earnings	0.80 (0.04) [0.51]	0.06 (0.12) [0.02]
Diversifying acquisitions	0.25 (0.06) [0.07]	0.04 (0.05) [0.00]
R&D	0.65 (0.05) [0.33]	0.09 (0.05) [0.02]
Advertising	0.62 (0.08) [0.02]	0.11 (0.06) [0.01]
SG&A	0.14 (0.01) [0.03]	0.08 (0.08) [0.02]
Return on assets	0.31 (0.07) [0.40]	0.02 (0.06) [0.01]
Operating return on assets	0.18 (0.03) [0.07]	0.03 (0.11) [0.00]

Notes: Each entry comes from a separate regression. Real data: regress the manager's average residual in the second firm on the average residual in the first firm. Placebo: use the second-firm residual measured three years before the manager arrives. Standard errors are in parentheses and R^2 in brackets.

Interpretation. The persistence coefficients are positive and sizable in real data but near zero in the placebo timing test. This supports the claim that manager-associated residual behavior travels with the manager across firms and is not purely driven by pre-existing trends in the destination firm.

10.6 Table VI: Size distribution of manager fixed effects

Table 6: Size distribution of manager fixed effects

	Median	Std. dev.	25th pct.	75th pct.
Investment	0.00	2.80	−0.09	0.11
Inv. to Q sensitivity	−0.02	0.66	−0.16	0.12
Inv. to CF sensitivity	0.04	1.01	−0.17	0.28
Number of acquisitions	−0.04	1.50	−0.54	0.41
Leverage	0.01	0.22	−0.05	0.09
Interest coverage	0.00	860.0	−56.0	51.7
Cash holdings	0.00	0.06	−0.03	0.02
Dividends/earnings	−0.01	0.59	−0.13	0.11
Diversifying acquisitions	−0.04	1.05	−0.28	0.21
R&D	0.00	0.04	−0.10	0.02
SG&A	0.00	0.66	−0.09	0.09
Advertising	0.00	0.04	−0.01	0.01
Return on assets	0.00	0.07	−0.03	0.03
Operating return on assets	0.00	0.08	−0.02	0.03

Notes: Fixed effects are taken from the most saturated specifications in Tables 3 and 4. Fixed effects are weighted by inverse standard errors to account for estimation noise.

Interpretation. This table turns “manager effects exist” into “manager effects are big enough to matter.” For example, moving from the 25th to 75th percentile manager in ROA fixed effects corresponds to about a 6 percentage point gap (from −0.03 to 0.03), which is economically meaningful for large firms.

10.7 Table VII: Relationship between manager fixed effects (styles)

Table 7: Relationship between manager fixed effects (each cell is a separate weighted regression coefficient; standard errors in parentheses)

	Investment	Inv. to Q	Inv. to CF	Cash hold.	Leverage	R&D	ROA
Investment							0.00 (0.00)
Inv. to Q sensitivity	**6.8** (0.92)						**0.03** (0.01)
Inv. to CF sensitivity	0.02 (0.6)	−0.23 (0.11)					−0.01 (0.01)
Cash holdings	−1.10 (1.62)	−0.79 (1.71)	−0.46 (1.72)				−0.12 (0.05)
Leverage	−0.39 (0.55)	−0.28 (0.59)	−0.63 (0.60)	−0.40 (0.17)			−0.02 (0.02)
R&D	**0.07** (0.00)	**0.08** (0.02)	−0.03 (0.01)	−0.23 (0.04)	−0.02 (0.01)		0.11 (0.11)
Advertising	0.01 (0.01)	**0.02** (0.01)	−0.01 (0.01)	−0.01 (0.04)	0.00 (0.01)	0.25 (0.15)	**0.31** (0.15)
Number of acquisitions	−0.27 (0.11)	0.08 (0.10)	**0.23** (0.10)	**0.01** (0.00)	**0.02** (0.01)	−0.01 (0.00)	−0.01 (0.00)
Diversifying acquisitions	−0.30 (0.13)	−0.14 (0.15)	0.14 (0.14)	0.01 (0.01)	0.01 (0.02)	−0.01 (0.00)	−0.01 (0.00)
SG&A	−0.22 (0.01)	−0.30 (0.04)	**0.10** (0.03)	0.54 (0.56)	0.06 (0.21)	−4.32 (0.90)	−3.36 (0.62)

Notes: Each entry is the slope coefficient from a weighted regression of the row manager fixed effects on the column manager fixed effects. Bold indicates statistical significance at the 10% level as highlighted in the paper.

Interpretation. The table reveals that manager heterogeneity is not one-dimensional. For example, investment aggressiveness, acquisition propensity, and cost structure move together in systematic ways, supporting the "management style" interpretation.

10.8 Table VIII: Governance, compensation, and manager fixed effects

Table 8: Governance, compensation, and manager fixed effects

	Blockholder share	Residual total comp	Residual salary comp
Return on assets	0.012 (0.006)	0.72 (0.24)	2.86 (0.57)
Investment	0.278 (0.252)	0.02 (0.01)	−0.08 (0.06)
Inv. to Q sensitivity	0.246 (0.053)	0.08 (0.03)	0.19 (0.13)
Inv. to CF sensitivity	−0.004 (0.088)	−0.06 (0.04)	−0.06 (0.07)
Cash holdings	−0.001 (0.007)	−0.02 (0.15)	−0.26 (0.29)
Leverage	−0.018 (0.021)	0.04 (0.26)	−0.01 (0.18)
R&D	0.009 (0.006)	−0.94 (0.08)	−0.33 (0.90)
Advertising	0.008 (0.007)	2.18 (0.93)	1.36 (0.54)
Number of acquisitions	−0.568 (0.131)	0.10 (0.05)	0.00 (0.03)
Diversifying acquisitions	−0.617 (0.092)	0.09 (0.04)	0.03 (0.05)
SG&A	−0.027 (0.093)	−0.16 (0.04)	−0.09 (0.25)

Notes: Column (1) regresses manager fixed effects on blockholder ownership concentration in the manager's second firm (weighted by estimation precision). Columns (2)–(3) relate manager fixed effects to manager-level residual compensation measures.

Interpretation. Managers with higher performance fixed effects (ROA) appear to be matched to more concentrated-ownership firms and also earn higher residual compensation, consistent with governance and pay mechanisms that select and reward performance-enhancing managerial styles.

10.9 Table IX: CEO birth cohort and MBA effects

Table 9: CEO birth cohort and MBA effects on firm policies

Dependent variable	Year of birth (×10)	MBA
(1) Investment	0.017 (0.005)	0.016 (0.010)
(2) Inv. to Q sensitivity	−0.013 (0.003)	0.017 (0.006)
(3) Inv. to CF sensitivity	0.118 (0.014)	−0.075 (0.026)
(4) Number of acquisitions	0.001 (0.037)	−0.017 (0.056)
(5) Leverage	0.024 (0.007)	0.011 (0.008)
(6) Interest coverage	−6.50 (2.67)	0.924 (3.41)
(7) Cash holdings	−0.005 (0.002)	−0.001 (0.003)
(8) Dividends/earnings	0.000 (0.003)	−0.009 (0.004)
(9) Diversifying acquisitions	−0.036 (0.015)	0.040 (0.017)
(10) R&D	−0.003 (0.002)	−0.002 (0.002)
(11) Advertising	−0.001 (0.002)	0.003 (0.003)
(12) SG&A	0.002 (0.003)	−0.004 (0.003)
(13) Return on assets	−0.003 (0.004)	0.012 (0.005)
(14) Operating return on assets	−0.002 (0.003)	0.008 (0.003)

Notes: Each row is a separate regression with firm and year fixed effects and CEO tenure controls. Standard errors are clustered at the manager level. Rows (2)–(3) come from a specification where birth year and MBA interact with Q and cash flow to isolate sensitivity effects.

Interpretation. Observable CEO characteristics explain part of the heterogeneity: younger cohorts tend to use more aggressive financial policies, while MBAs show distinctive investment sensitivity patterns (more Q-driven and less cash-flow-driven investment), consistent with the "MBA style" narrative.

11 What to remember

1. The empirical strategy is a **manager switchers design**: identify manager fixed effects using executives observed in multiple firms.

2. The benchmark regression (1) is **not causal** in the sense of randomized assignment; it documents systematic co-movement between manager identity and firm outcomes.

3. Manager effects are found for **policies and performance**, they are **persistent across firms**, and they form **coherent cross-policy patterns** (styles).

4. Governance and compensation correlate with style/performance measures, suggesting potential efficiency implications.

Final summary (what the paper does, and what it finds)

Research question. The paper asks whether individual top managers (CEOs and other executives) systematically shape firm policies and performance, above and beyond persistent firm characteristics and common time shocks. The core empirical idea is to treat "style" as a manager-specific component that persists across jobs and therefore can be detected when the same manager appears in multiple firms over time.

Empirical strategy. The authors build a manager–firm matched panel that tracks top executives across firms and over time, enabling regressions with firm fixed effects and year fixed effects plus manager fixed effects. Intuitively, firm fixed effects absorb time-invariant firm heterogeneity, year fixed effects absorb aggregate shocks, and manager fixed effects capture persistent differences across managers that reappear when the manager moves to a new firm.

Main results: managers have economically meaningful fixed effects. Adding manager fixed effects helps explain a nontrivial share of the remaining within-firm variation in many corporate outcomes. Manager fixed effects are particularly salient for some policies (e.g., acquisition/diversification activity, payout policy, interest coverage, and cost-cutting proxies), and different types of executives matter more for different margins (e.g., CFOs for some financial outcomes).

From fixed effects to "styles." Rather than only documenting that managers matter, the paper **shows that manager fixed effects across different corporate variables covary in interpretable ways.** This reveals systematic differences in managerial approaches, such as (i) internal vs. external growth orientation and (ii) degrees of "financial aggressiveness," with some managers consistently engaging more in acquisitions/diversification and displaying different investment and innovation patterns.

Performance, governance, and compensation. The paper also finds that managerial heterogeneity extends to performance measures (e.g., accounting returns), and that managerial styles correlate with performance fixed effects. Moreover, managers with higher performance fixed effects receive higher compensation, and they are more likely to be in firms with more concentrated ownership (a governance proxy), consistent with an environment where governance and managerial selection interact.

Observable traits linked to style. Finally, the paper relates part of the estimated heterogeneity to observable manager characteristics, emphasizing patterns for birth cohort (age) and MBA education: older cohorts appear, on average, more conservative, while MBAs appear, on average, more aggressive (and MBA status is positively related to performance in the reported accounting measures).

Economic intuition (why the findings make sense)

Intuition 1: "Style" as a portable decision rule. If managers carry portable preferences, beliefs, or heuristics (e.g., expansionary vs. conservative attitudes), then when they move across firms they may implement similar policy choices conditional on the environment. This portability implies that, after controlling for firm and year effects, a manager-specific residual component should still predict policies in the new firm.

Intuition 2: Where discretion is larger, style should be more visible. Manager effects should be more pronounced in decisions that are (i) complex, (ii) hard to contract on, or (iii) more discretion-intensive (e.g., M&A, diversification, payout/cash policies, and internal organization). This aligns with the paper's emphasis that the largest manager fixed effects show up in exactly these types of outcomes.

Intuition 3: Performance links to selection and governance. If some styles are systematically more performance-enhancing (rather than purely "different but equally optimal"), then:

- such styles should correlate with performance fixed effects,
- better governed firms should be more likely to employ these managers, and
- compensation should reflect a premium for managers associated with better performance.

The paper presents evidence consistent with these implications, while acknowledging that fully separating "manager-imposed style" from "optimal matching" is difficult.

Conclusion (what the paper concludes, in the authors' framing)

Core conclusion. The paper's primary objective is to document systematic behavioral differences across managers, and it concludes that there is substantial heterogeneity in investment, financing, and organizational strategies that depends on the identity of the executives in charge.

Scope and limitation stated by the paper. The authors emphasize that their framework does *not* estimate the causal effect of managers on firm policies or performance. Instead, it provides a simple and intuitive way to address important selection concerns when studying manager influence using observational data.

Broader takeaway. The paper positions manager heterogeneity as a first-order empirical object for corporate finance and organizational economics: a meaningful share of cross-firm and within-firm policy variation cannot be fully understood without incorporating a manager dimension, and some of this heterogeneity is partially predictable from observable traits such as cohort and MBA education.